

Lecture 13: Process Scheduling Policies

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Last Lecture

```
void scheduler() {
    while(true) {
        lock(process_table);
        foreach(Process p: scheduling_algorithm(process_table)) {
            if(p->state != READY) {
                continue;
            }
            p->state = RUNNING;
            unlock(process_table);
            swtch(scheduler_process, p);
            // p is done for now..
            lock(process_table);
        }
        unlock(process_table);
    }
}
```

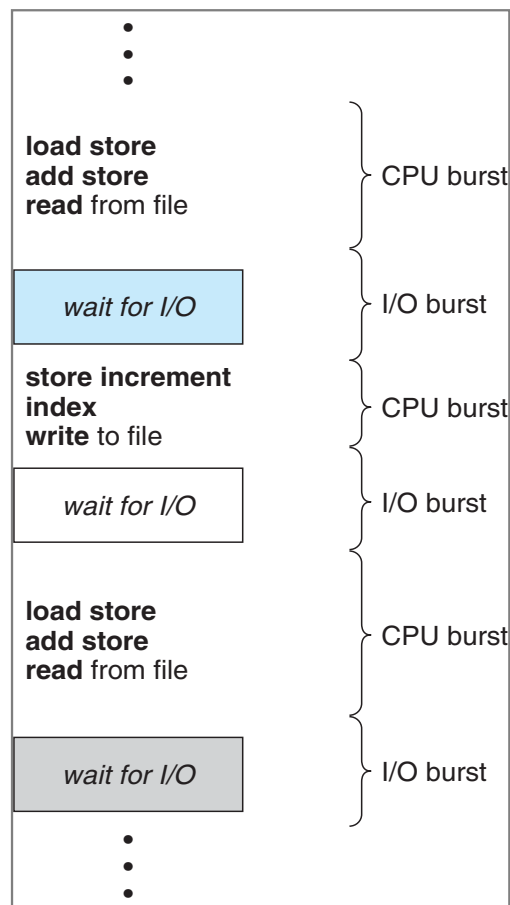
Today's Class

- **Process scheduling policies**

- First In First Out / First Come First Serve
- Shortest Job First
- Shortest Time to Completion First
- Round Robin
- Multi-level feedback scheduling
- Linux Completely Fair Scheduler (CFS)

Acknowledgements: Lecture content derived from the CS162 course offered at University of California, Berkeley

CPU and I/O Bursts



- Process execution model: programs alternate between bursts of CPU and I/O
 - Program typically uses the CPU for some period of time, then does I/O, then uses CPU again
 - Each scheduling algorithm is about which process to give to the CPU for use by its next CPU burst
 - With time slicing, process may be forced to give up CPU before finishing current CPU burst

Key Scheduling Goals

- **Response Time**

- What user sees: time to echo a keystroke in editor

- **Throughput**

- Total operations per second
 - Affected by the overheads of context switching

- **Fairness**

- Conflicts with the response time

- **Starvation**

- Due to improper resource allocation policy

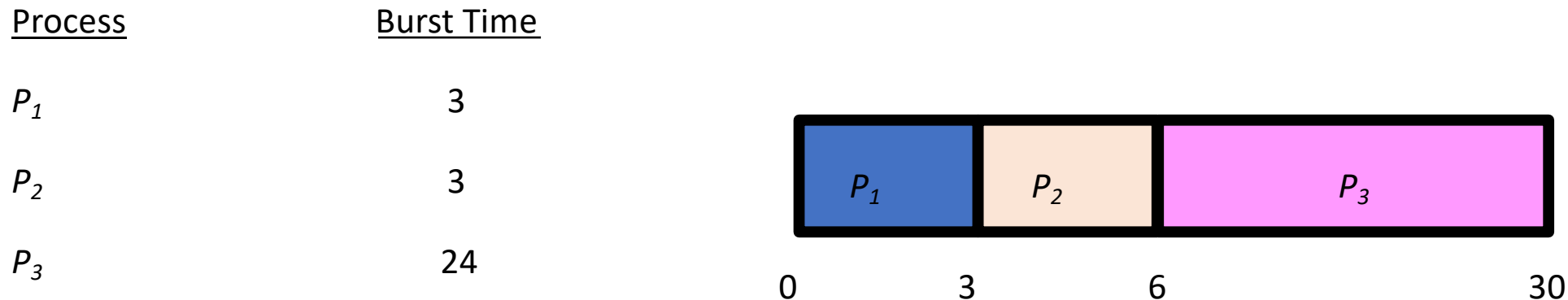
First-Come, First-Served (FCFS)

Run tasks in order of arrival.

Run task until completion (or blocks on IO).
No preemption

Also called FIFO

First-Come, First-Served (FCFS)



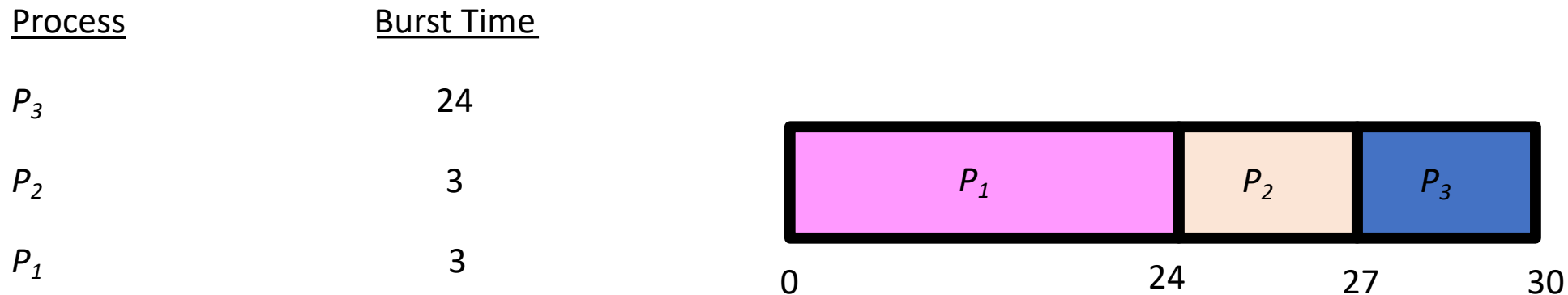
What is the average completion time?

$$\left(\frac{3+6+30}{3} = 13 \right)$$

What is the average waiting time?

$$\left(\frac{0+3+6}{3} = 3 \right)$$

First-Come, First-Served (FCFS)



What is the average completion time?

$$\left(\frac{24+27+30}{3} = 27 \right)$$

[v/s **13** earlier]

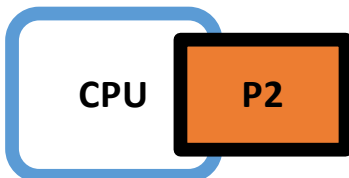
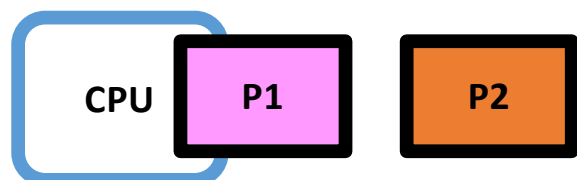
What is the average waiting time?

$$\left(\frac{0+24+27}{3} = 17 \right)$$

[v/s **3** earlier]

Can there be Convoy Effect in FCFS?

Convoy effect: Short process stuck behind long process



FIFO/FCFS very sensitive to arrival order
Lots of small tasks build up behind long tasks
FIFO is *non-preemptible*



Can FIFO lead to starvation?

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Shortest Job First

How can we minimize **average completion time**?

By scheduling jobs in order of
estimated completion time

Shortest Job First

Process

Burst Time

P_1

3

P_2

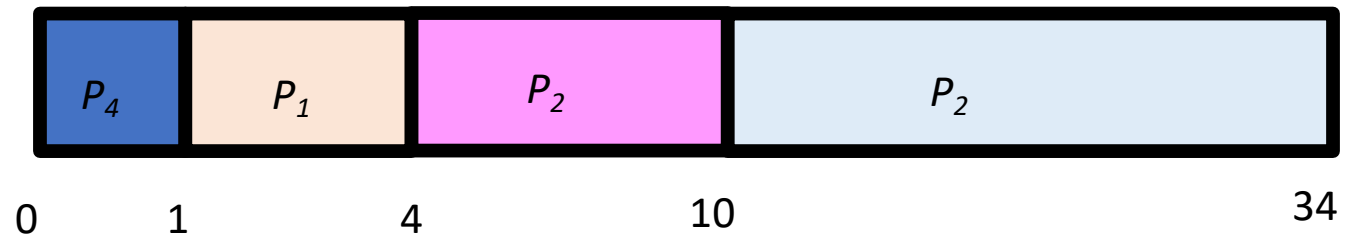
6

P_3

24

P_4

1

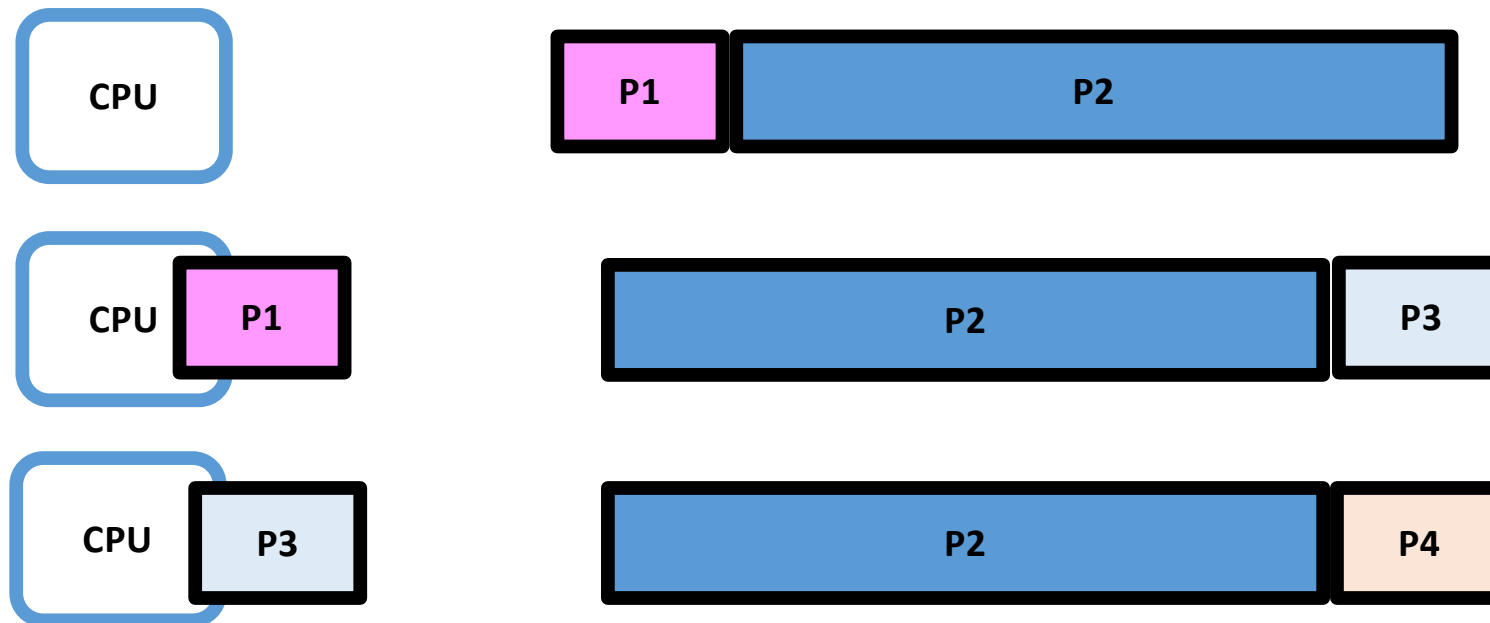


What is the average completion time?

$$\left(\frac{1+4+10+34}{4} = 12.25 \right)$$

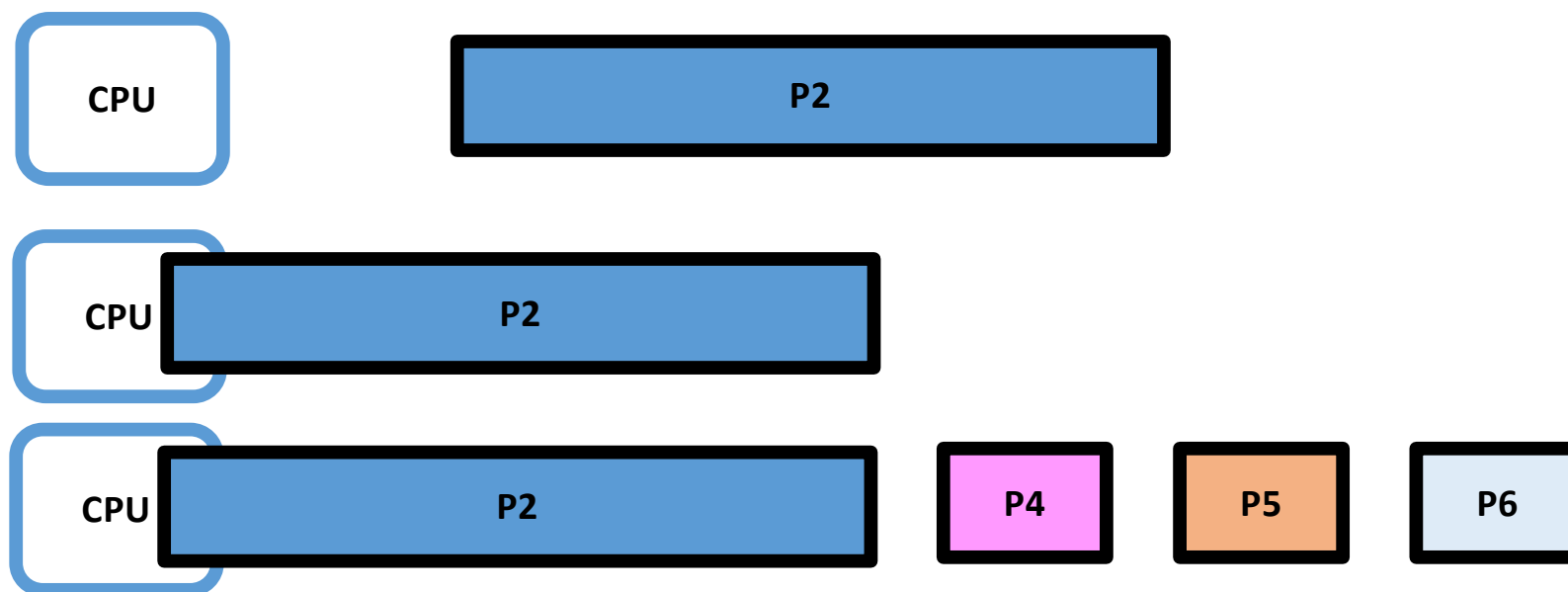
Can SJF Lead to Starvation?

Any scheduling policy that always favours a **fixed property** for scheduling leads starvation



Can there be Convoy Effect in SJF?

Any **non-preemptible** scheduling policy suffers from convoy effect



Summary

Property	FCFS	SJF
Optimize Average Completion Time	✗	✓
Prevent Starvation	✓	✗
Prevent Convoy Effect	✗	✗
Psychic Skills NOT Needed	✓	✗

If jobs arrive simultaneously

Any scheduling policy that always favors a fixed property for scheduling leads to starvation

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Shortest Time to Completion First (STCF)

Introduce the notion of **preemption**

A running task can be de-scheduled before completion.

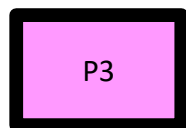
STCF

Schedule the task with the **least amount of time left**

Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	3	10
P_2	6	1
P_3	24	0
P_4	16	18

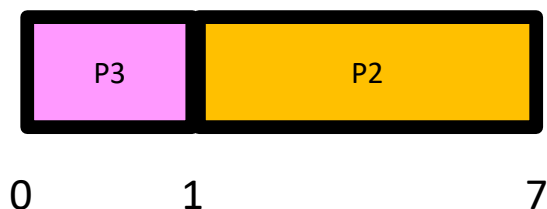


0 1

Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

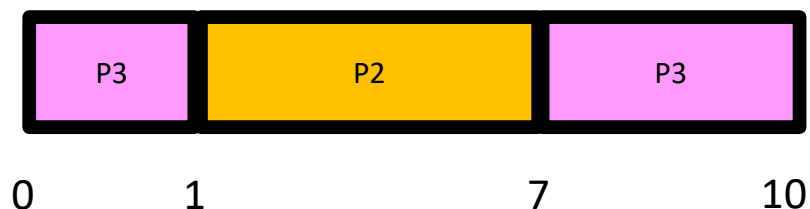
<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	3	10
P_2	6	1
P_3	23	0
P_4	16	18



Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

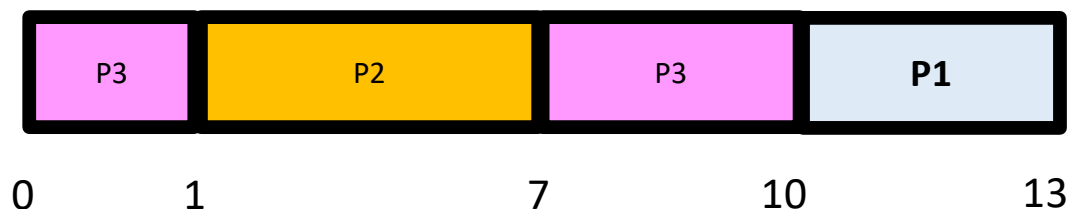
<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	3	10
P_2	0	1
P_3	23	0
P_4	16	20



Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

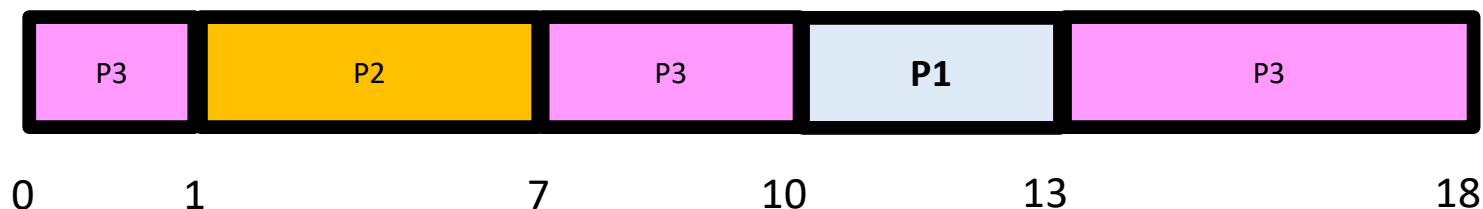
<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	3	10
P_2	0	1
P_3	20	0
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Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

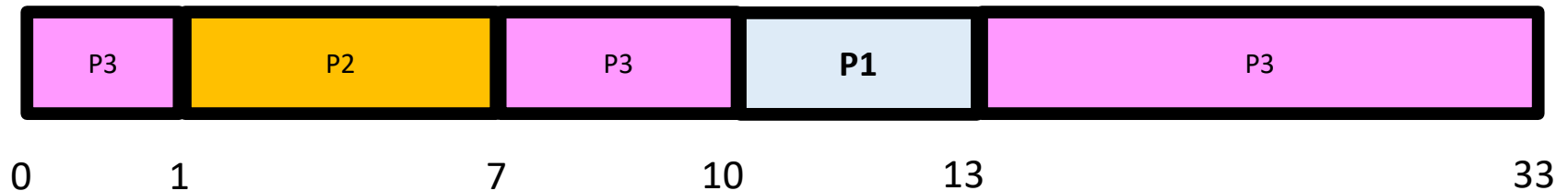
<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	0	10
P_2	0	1
P_3	15	0
P_4	16	18



Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

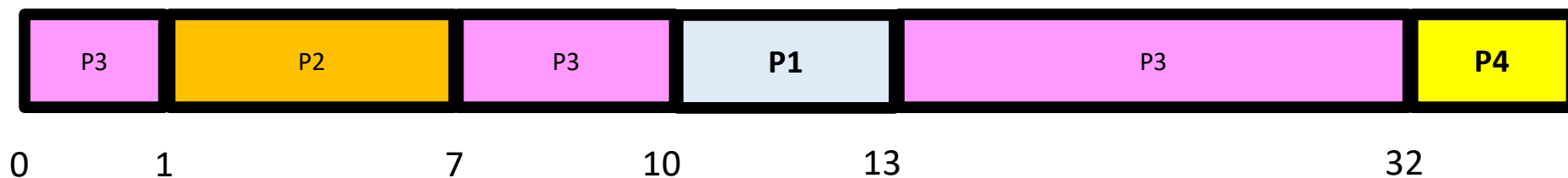
<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	0	10
P_2	0	1
P_3	0	0
P_4	15	18



Shortest Time to Completion First (STCF)

Schedule the task with the least amount of time left

<u>Process</u>	<u>Burst Time (left)</u>	<u>Arrival Time</u>
P_1	0	10
P_2	0	1
P_3	0	0
P_4	15	18



Summary

Property	FCFS	SJF	STCF
Optimize Average Completion Time	✗	✓	✓
Prevent Starvation	✓	✗	✗
Prevent Convoy Effect	✗	✗	✓
Psychic Skills NOT Needed	✓	✗	✗

If jobs arrive simultaneously

Always

Preemptible

Process execution time to be known

Any scheduling policy that always favors a fixed property for scheduling leads to starvation

Can we design a non-psycho, starvation-free scheduler with good response time?

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Round-Robin Scheduling

RR runs a job for a **time slice**
(a **scheduling quantum**)

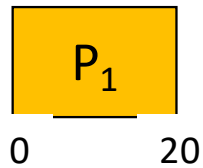
Once time slice over,
Switch to next job in ready queue.
=> Called **time-slicing**

RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	53
P_2	8
P_3	68
P_4	24

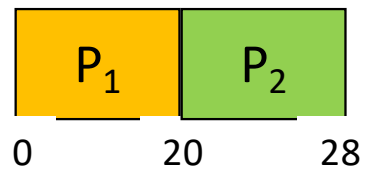
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	53 => 33
P_2	8
P_3	68
P_4	24



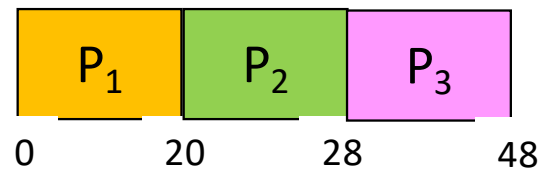
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	33
P_2	8 => 0
P_3	68
P_4	24



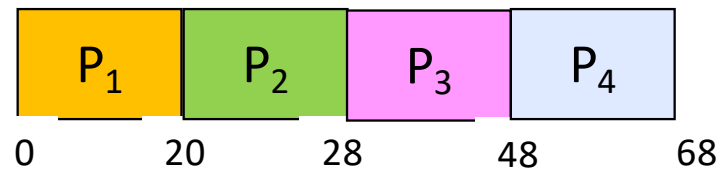
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	33
P_2	0
P_3	68 => 48
P_4	24



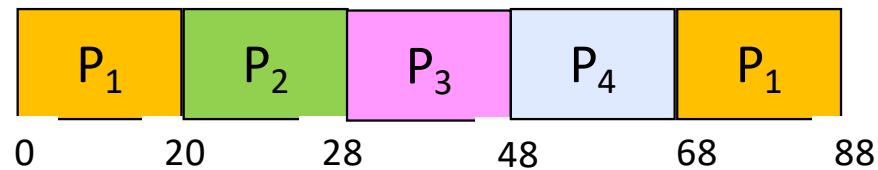
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	33
P_2	0
P_3	48
P_4	24 => 4



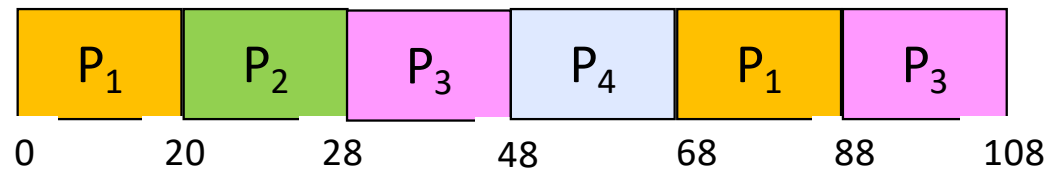
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	33 => 13
P_2	0
P_3	48
P_4	4



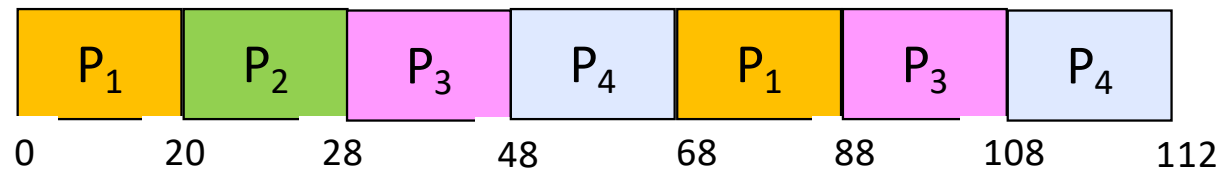
RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	13
P_2	0
P_3	48 $\Rightarrow$ 28
P_4	4



RR with Time Quantum = 20

<u>Process</u>	<u>Burst Time</u>
P_1	13
P_2	0
P_3	28
P_4	4 $\Rightarrow$ 0



RR with Time Quantum = 20

Waiting time

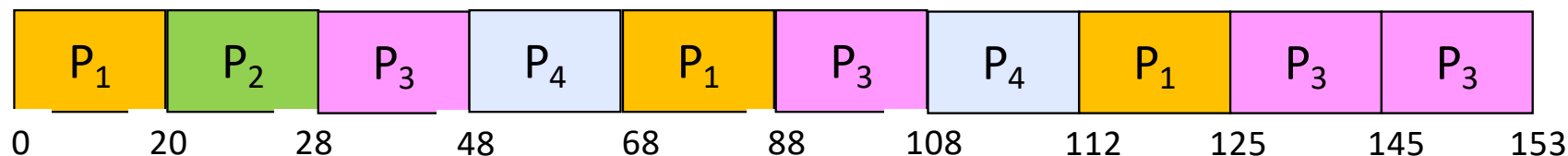
- $P_1 = 0 + (68-20) + (112-88) = 72$
- $P_2 = (20-0) = 20$
- $P_3 = (28-0) + (88-48) + (125-108) + 0 = 85$
- $P_4 = (48-0) + (108-68) = 88$

Average waiting time

$$\left(\frac{72+20+85+88}{4} = 66.25 \right)$$

Average completion time

$$\left(\frac{125+28+153+112}{4} = 104.25 \right)$$



Summary

Property	FCFS	SJF	STCF	RR
Optimize Average Completion Time	✗	✓	✓	✗
Optimize Average Waiting Time	✗	✗	✗	✓
Prevent Starvation	✓	✗	✗	✓
Prevent Convoy Effect	✗	✗	✓	✓
Psychic Skills NOT Needed	✓	✗	✗	✓

High in case of long running processes

Doesn't favor any fixed property

Due to time slicing

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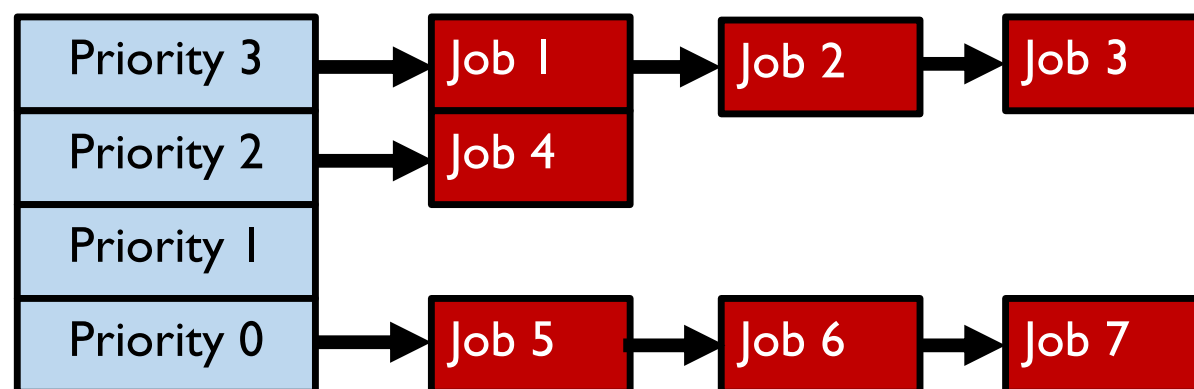
What we Want?

It cannot be achieved without having psychic skills

Would it be true for IO/interactive jobs having short CPU bursts?

Property	STCF	RR
Optimize Average Completion Time	✓	✗
Optimize Average Waiting Time	✗	✓
Prevent Starvation	✗	✓
Prevent Convoy Effect	✓	✓
Psychic Skills NOT Needed	✗	✓

Approximation of CPU Bursts using Priority Scheduling



- Priority scheduling always run the **ready** process (or job) with *highest priority*
- Systems may try to set priorities according to some **policy goal**
 - Example: Give IO/interactive jobs higher priority than long calculation
- How to achieve fairness and avoid starvation?
 - Elevate priority of threads that don't get CPU time

Multi-level Feedback Queue

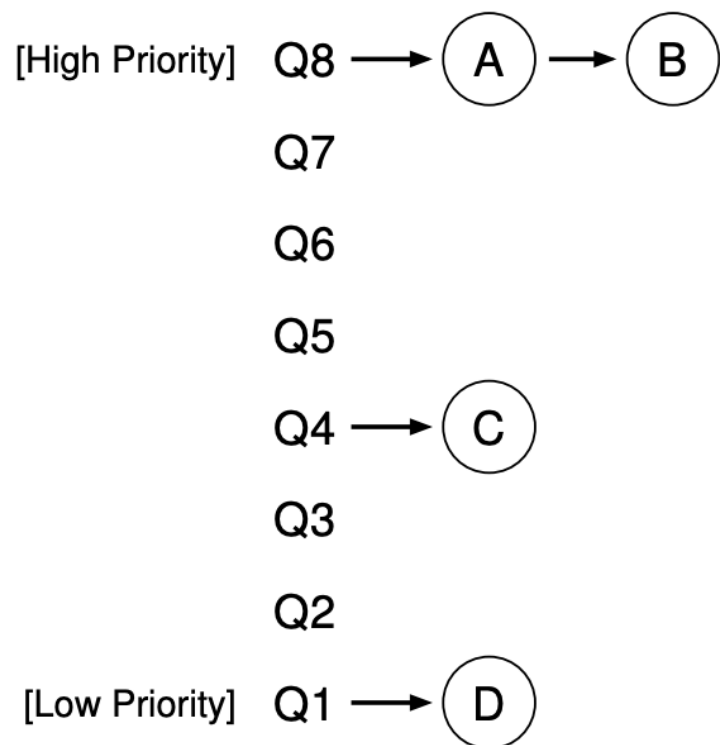


Figure 8.1: MLFQ Example

- Create distinct **queues for ready processes**, each assigned a different **priority level**
- All processes belong to any one queue at any given time
- Processes can move between queues
- Use priorities to decide from which queue process should be picked next
- Individual queues run RR with increasing time slice

MLFQ (Naïve Version)

Rule 1

If $\text{Priority}(A) > \text{Priority}(B)$ (different queues)
A runs (B doesn't).

Rule 2

If $\text{Priority}(A) = \text{Priority}(B)$, A & B run in RR.

Key question:

How do you set the priorities?

Vary the priority of a job based on its *observed behavior*

Learning behavior

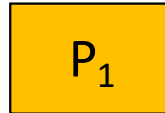
Rule 3

When a job enters the system, it is placed at the highest priority (the topmost queue).

Rule 4a

If a job uses up an entire time slice while running, its priority is *reduced* (i.e., it moves down one queue).

Learning behavior



Long running CPU-bound

$q = 10$ ms



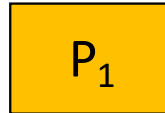
$q = 5$ ms



$q = 2$ ms



Learning behavior



Long running CPU-bound

$q = 10$ ms



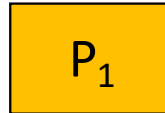
$q = 5$ ms



$q = 2$ ms



Learning behavior



Long running CPU-bound

$q = 10$ ms



$q = 5$ ms



$q = 2$ ms



Learning behavior

P_1

Long running CPU-bound

$q = 10$ ms



$q = 5$ ms



$q = 2$ ms



Learning behavior

Rule 3

When a job enters the system, it is placed at the highest priority (the topmost queue).

Rule 4a

If a job uses up an entire time slice while running, its priority is *reduced* (i.e., it moves down one queue).

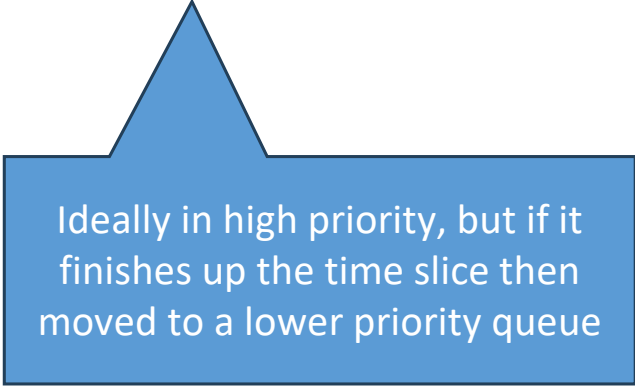
Rule 4b

If a job gives up the CPU before the time slice is up, it stays at the *same* priority level.

Learning behavior

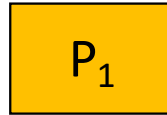
Where do IO-bound/interactive jobs end up?

a) High Priority Queue b) Low Priority Queue



Ideally in high priority, but if it finishes up the time slice then moved to a lower priority queue

Learning behavior



Long running CPU-bound

$q = 10$ ms



$q = 5$ ms

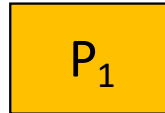


$q = 2$ ms



Schedule

Learning behavior



Long running CPU-bound

$q = 10$ ms



$q = 5$ ms

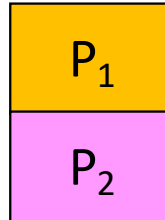


$q = 2$ ms



Schedule

Learning behavior



Long running CPU-bound

Computes for 1 ms and then IO for 1 ms

$q = 10$ ms



$q = 5$ ms

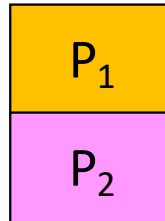


$q = 2$ ms



Schedule

Learning behavior



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$q = 5$ ms

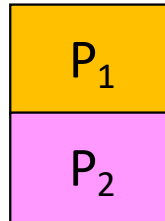


$q = 2$ ms



Schedule

Learning behavior



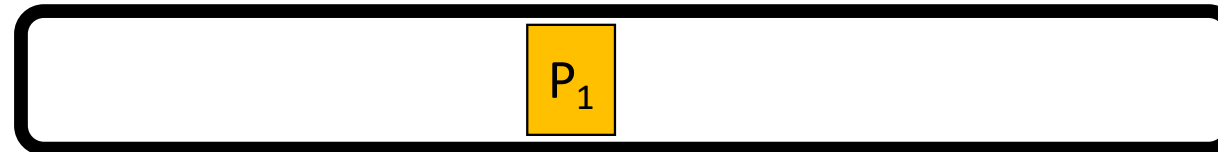
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$q = 5$ ms

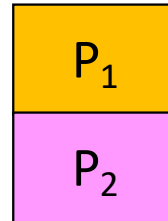


$q = 2$ ms



Schedule

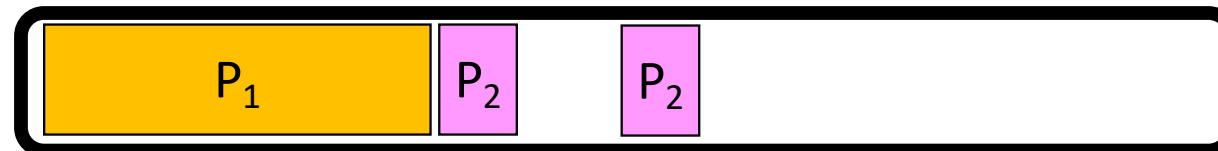
Learning behavior



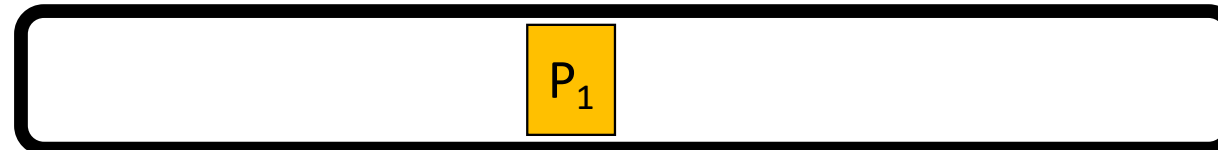
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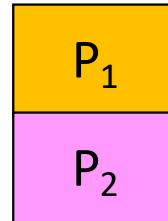


$q = 2$ ms



Schedule

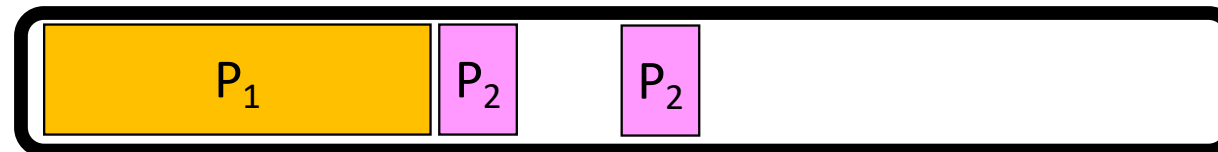
Learning behavior



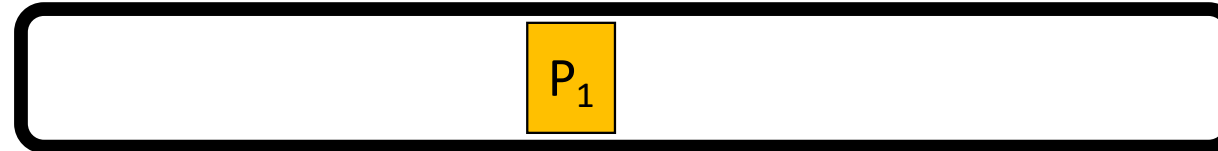
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$q = 5$ ms

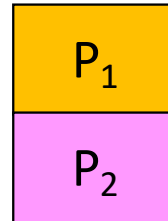


$q = 2$ ms



Schedule

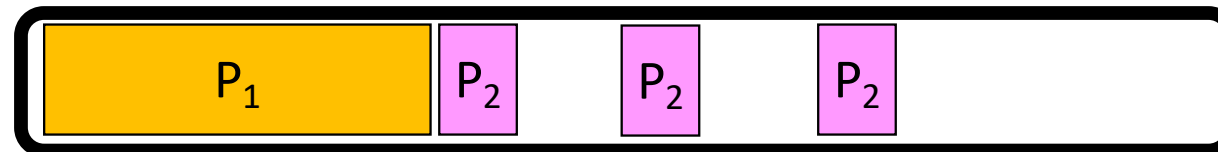
Learning behavior



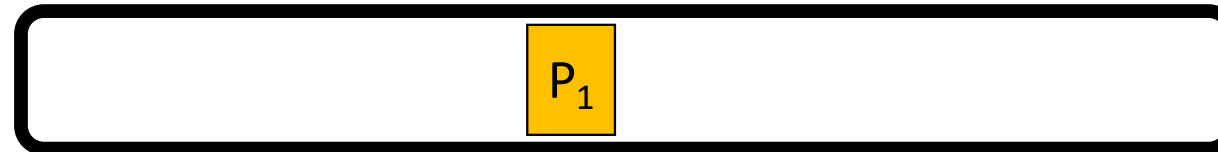
Long running CPU-bound

Computes for 1 ms and then IO for 1 ms

$q = 10$ ms



$q = 5$ ms

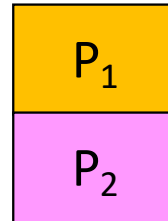


$q = 2$ ms



Schedule

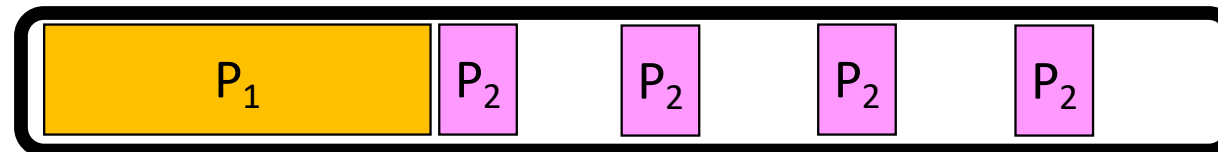
Learning behavior



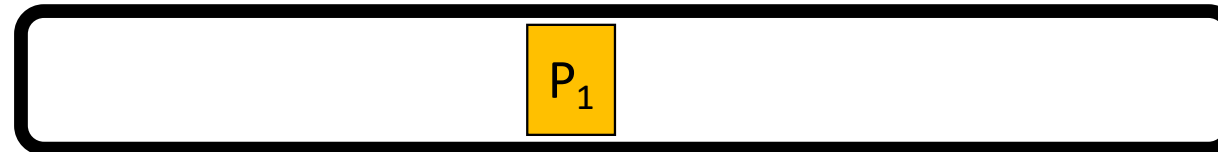
Long running CPU-bound

Computes for 1 ms and then IO for 1 ms

$q = 10$ ms



$q = 5$ ms



$q = 2$ ms



Schedule

MLFQ (Naïve Version): Pseudocode

```
foreach(queue : PrioritizedReadyQueues) { // RR
    foreach(job : queue) { // RR
        int timeslice = queue.getSlice();
        // Execute job for timeslice associated with this queue
        // If job used full timeslice then push job in
        // ready queue with one lower priority level
    }
}
```

Are we done?

Can MLFQ can be gamed?

Intentionally insert IO request just before time quanta to stay on queue.

MLFQ is subject to starvation?

Yes!

Are we done?

MLQF can be gamed:

Intentionally insert IO request just before time quanta to stay on queue.

MLQF is subject to starvation:

Systematically prioritize higher-priority queues

Rule 4

Once a job uses up its time allotment at a given levels (regardless of how many times gave up CPU), reduce priority

Rule 5

After some time period S , move all jobs in system to the topmost queue.

Today's Class

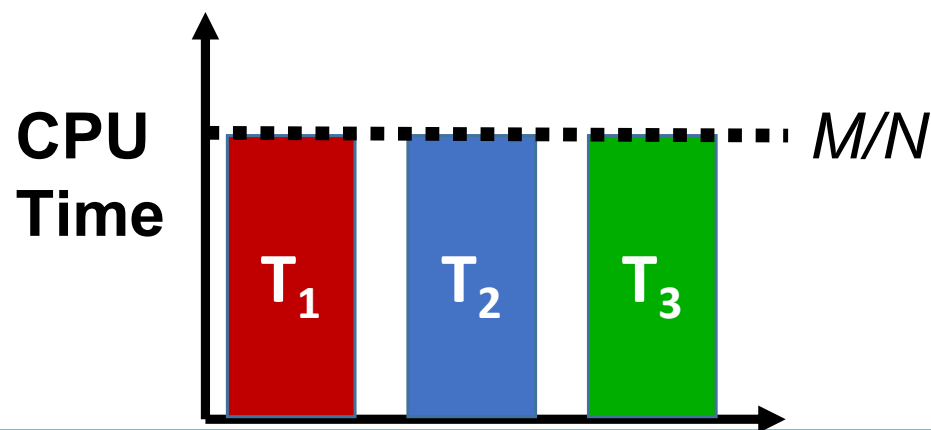
- **Process scheduling policies**

- First In First Out / First Come First Serve
- Shortest Job First
- Shortest Time to Completion First
- Round Robin
- Multi-level feedback scheduling
- Linux Completely Fair Scheduler (CFS)

Acknowledgements: Lecture content derived from the CS162 course offered at University of California, Berkeley

Linux Completely Fair Scheduler

- Goal: Each process gets an equal share of CPU
- Attempts to divide the CPU time fairly (equally) among all the processes
 - If the latency to resume a scheduler is M units of time, N processes will get M/N time slice of CPU



Linux CFS

- **Constraint 1:** *Scheduling Latency*

- Target Latency: 20ms, 4 Processes

- Each process gets 5ms time slice

- Target Latency: 20 ms, 200 Processes

- Each process gets **0.1ms** time slice
- Recall Round-Robin: Huge context switching overhead

Linux CFS

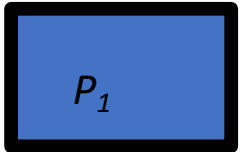
- **Constraint 2:** *Minimum Granularity for scheduling latency*
- Target Latency 20ms, Minimum Granularity 1ms, 100 processes
 - Each process gets 1ms time slice

Linux CFS

- Which process to pick from the ready queue?
- **Constraint 3:** *Pick the process from the ready queue with minimum vruntime (helps the IO/interactive jobs)*
 - Uses virtual runtime (vruntime) variable in PCB to keep track of time a process has executed on the CPU, and it is updated at every **context switch**
 - $\text{vruntime} += t * \text{weight}$ based on process priority

Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	5
P_2	10
P_3	16
P_4	16

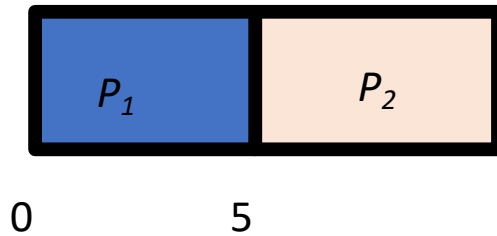


0

- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$

Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	10
P_3	16
P_4	16

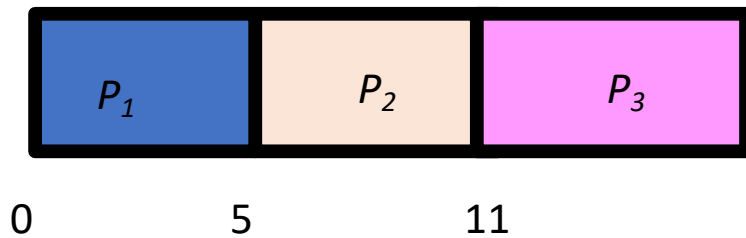


- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$

Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	4
P_3	16
P_4	16

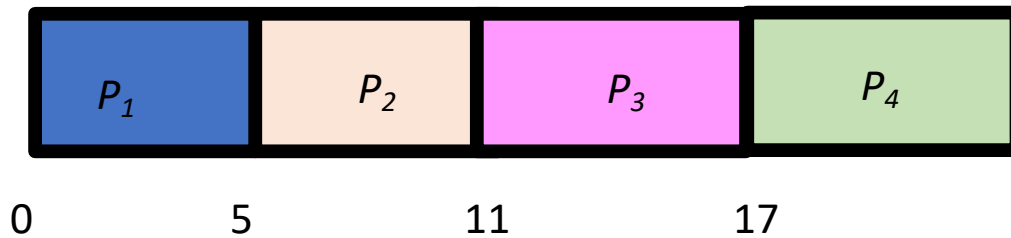
- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$



Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	4
P_3	10
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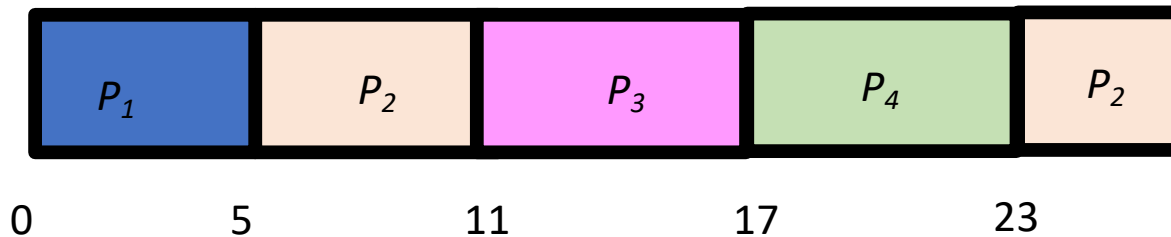
- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$



Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	4
P_3	10
P_4	10

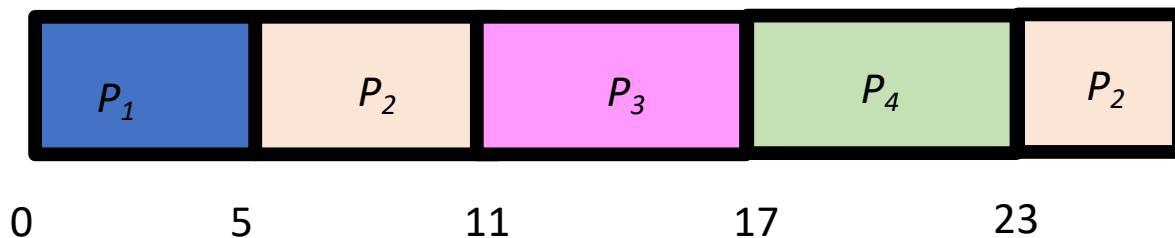
- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$



Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	0
P_3	10
P_4	10

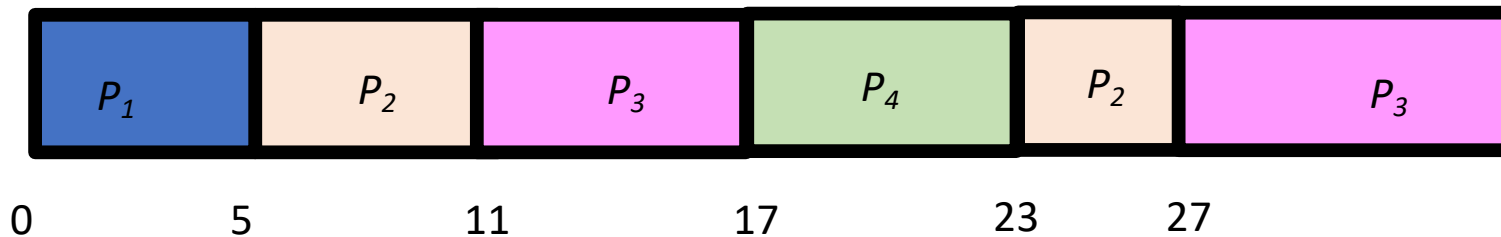
- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$
- Time slice after P_2 terminates = $20/2 = 10\text{ms}$



Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	0
P_3	0
P_4	10

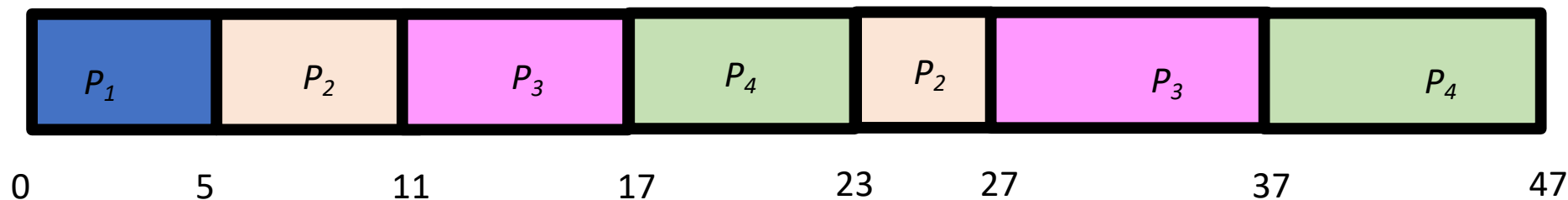
- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
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- Time slice after P_2 terminates = $20/2 = 10\text{ms}$



Linux CFS

<u>Process</u>	<u>Burst Time</u>
P_1	0
P_2	0
P_3	0
P_4	0

- Scheduling latency is **20ms**, minimum granularity is 1ms, and weight=1
- Time slice in beginning = $20/4 = 5\text{ms}$
- Time slice after P_1 terminates = $20/3 \sim 6\text{ms}$
- Time slice after P_2 terminates = $20/2 = 10\text{ms}$



Next Lecture

- Dynamic memory allocation
- Assignment-03 will be released today with the deadline of 12th October (there will no extensions)